6.8371 Problem Set 9 Write-Up

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12/5/2022

Motivation:

I chose to do the seam carving assignment (a problem set from a previous year). I thought it would be interesting to work on this assignment because I had heard of seam carving before, and I roughly knew what it was, but I never learned how it worked. This problem set provided me with the opportunity to implement this technique from scratch so that I could understand it technically and explore its applications.

Background:

In many image applications, it is desirable to be able to resize the dimensions of an image, while retaining its most salient features. There has been significant prior work in developing such content-aware image resizing algorithms. One such algorithm, proposed by Suh et al. [2], first uses a saliency map algorithm to identify the most salient parts of an image. They authors construct a cropping method to crop an image to a smaller size while retaining as many of its salient elements as possible. Suh et al. also explore the use of a facial detector, which can be used in place of the saliency map when determining the appropriate crop. Efros and Freeman [3] also propose a method to resize images with a focus on maintaining the image's "texture." Specifically, their method called "image quilting" stitches together small parts of an image in a natural way to form a much larger texture. This enables them to expand images of much larger sizes (e.g., turn an image of a few apples into an image of hundreds of apples). Setlur et al. [4] approach the image resizing problem from the perspective of mobile devices; they seek to find a way to reduce the resolution and size of images meant for larger displays, so that they can be displayed on phones and tablets. Their approach involves identifying the most important objects in an image and removing them. The "background" is then resized before the important objects are reinserted. Their method is designed to work well even when multiple objects are present in the image.

Finally, Avidan and Shamir [1] propose seam carving, another method to resize images by removing (or adding) horizontal and vertical "seams" from (to) the image (see examples of seams in Figure 1). I describe the process in more detail below.



Figure 1: Example of a vertical and horizontal seam.

Algorithm:

Seam carving is an image resizing technique that involves iteratively removing "seams" from an image. A horizontal seam is defined as a set of pixels in an image, such that (1) there is one pixel in each row and (2) the vertical distance between any two horizontally adjacent pixels in the set is no more than one. Qualitatively, the set of pixels form a connected region from the left side of the region to the right, resembling a "seam" (see Figure 1). Vertical seams are defined analogously.

To determine which seams to remove, we need a measure of how salient each pixel is. In the authors' words, we assign each pixel an "energy" level. To compute this energy value, at each pixel, we compute the image's gradient with respect to both the x- and y-directions. These can be computed using the gradientX() and gradientY() function, which we used in the Sobel filter. Then, for each pixel, we compute the magnitude of these gradients across channels and across both the x- and y-directions. The authors use an L1 magnitude, which we also adopt in this implementation. Since larger gradients correspond to more salient, edge-like features, we'd like to remove seams that have the lowest total energy levels.

Specifically, to reduce a W x H sized image's height by h pixels, we proceed as follows. We first identify the horizontal seam in the image whose pixels' total energy level is smallest. This can be computed using a dynamic programming algorithm, in which we traverse the image column by column; for each pixel in column c, we compute only the minimum attainable energy sum for a seam that starts at the first column and ends in column c. Then, the minimum energy seam can be recovered by backwards-traversing the minimum energy sums from right to left. Once we have identified this seam, we remove it by creating a W x (H - 1) sized image, where the pixels in the original image below the missing seam are shifted upwards. We repeat this process on the new image to remove another row of pixels. Repeating this procedure h times yields us a shorter W x (H - h) image.

To reduce both the height and the width of the image, seam carving can be applied in horizontal direction first (to reduce the height), and be applied in the vertical direction afterwards (to reduce the width).

I also implement a variant of seam carving that enables us to expand the dimension of an image, rather than just shrink it. In particular, in order to increase the height of a H x W image by h pixels, we determine the h horizontal seams we *would have* removed if we were to shrink the height by h pixels. These seams roughly correspond to the pixels that are least important (and thus would have been removed first if the image had been taller). We duplicate these h seams, which increases the height to H + h. As before, we can change both the height and the width by using seams in the vertical direction as well.

Implementation and Results

I first implemented a function to compute the energy level at each (x, y)-coordinate, where higher energy levels correspond to parts of the image that we would like to retain. A normalized plot of the energy levels is shown in Figure 2, which corresponds to the Boston skyline picture in Figure 1.



Figure 2: Normalized energy levels.

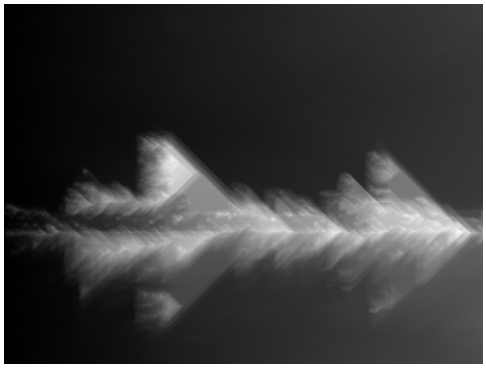


Figure 3: Normalized minimum cumulative energy levels for horizontal seams.

Next, I implemented a function to compute the minimum cumulative energy levels (as described above), which is used in the dynamic programming computation of the seam with the minimum energy level. The normalized cumulative energy levels for the horizontal seams (for the same image) are shown in Figure 3. The key point here is that as we move from left to right, the energy levels rise sharply (the image becomes much brighter) as we encounter the salient regions of the image (i.e., the buildings).

Using these cumulative energy values, we can work backwards from right to left to determine the pixel locations of the horizontal seam with the smallest energy sum. We can do the same in the vertical direction. Figure 1 shows the lowest-energy seams in both directions. As you can see, they do not overlap with the buildings, but rather touch the more "unimportant," background pixels.

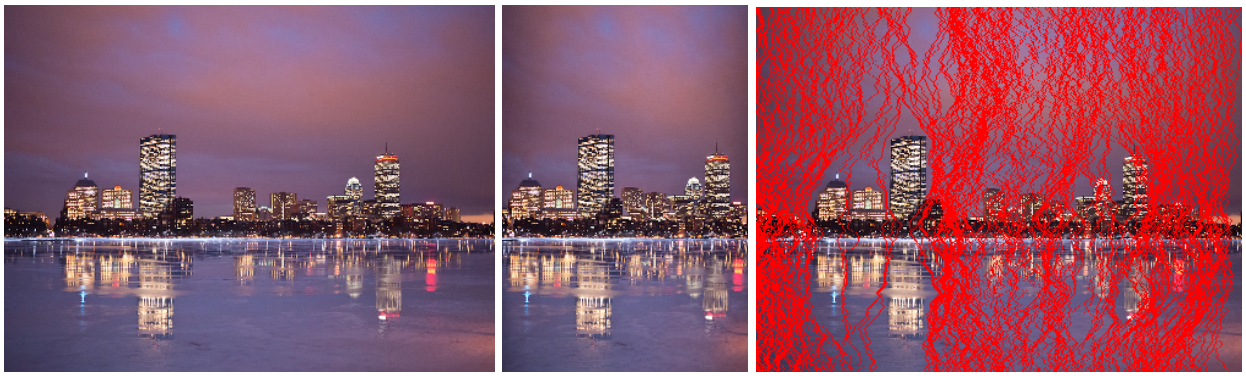


Figure 4: Original image (left), result of reducing width by half (middle), carved pixels (right).



Figure 5: Original image (left), result of reducing width by half (middle), result of reducing both dimensions by half (right).

The middle image in Figure 4 shows the result of seam carving out half of the pixels length-wise. We can see that the prominent Boston buildings are still retained in the image, and that the carved seams (in red) mostly remove the sky and water. Figure 5 shows seam carving applied to two birds, with the rightmost example carving both dimensions by half. As desired, the birds generally retain their original appearance. These images, which contain roughly 100-200k pixels, take around 10 seconds to generate.

We also implement an extension of seam carving (described earlier) that enables us to increase the dimensions of images. This involves implementing a function that determines which seams to duplicate (using seam carving), as well as a function that resizes the image given a seam to duplicate. We plot the result below for an elephant image in Figure 6. We can see that the primary objects in the image (the elephant and the tree stump) look the same in the enlarged image, as desired, but that the background is lengthened.

Overall, implementing seam carving was very exciting. The most challenging parts were figuring out how to project the seams computed from successively smaller images onto a single image of fixed size (e.g., right image in Figure 4). A similar technique was also necessary when computing the enlarged image via seam duplication (e.g., left image of Figure 6). It was a fun last problem set, and a great way to finish off the semester!



Figure 6: Original image (left), result of increasing width by half (right).

References:

[1] Avidan, Shai, and Ariel Shamir. "Seam Carving for Content-Aware Image Resizing." *ACM SIGGRAPH 2007 Papers.* 2007.

[2] Suh, Bongwon, et al. "Automatic Thumbnail Cropping and Its Effectiveness." *Proceedings of the 16th Annual ACM Symposium on User Interface Software and Technology.* 2003.

[3] Efros, Alexei A., and William T. Freeman. "Image Quilting for Texture Synthesis and Transfer." *Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques.* 2001.

[4] Setlur, Vidya, et al. "Automatic Image Retargeting." *Proceedings of the 4th International Conference on Mobile and Ubiquitous Multimedia.* 2005.

Ethics Response

There are several ways I could manipulate images that might make people look more "attractive." From an image quality standpoint, I would propose to make blurry or noisy images higher quality by using image sharpening and denoising techniques. If the image orientation is tilted, I would propose applying rotations to make the image look more professional. If the subject has blemishes on the skin (e.g., acne) or marks on their clothing (e.g., stains) that they do not wish to emphasize, I would propose applying techniques to decrease the visibility of those areas. I may also try to emphasize features of the

individuals that are most appealing. For instance, people's eyes are often seen as quite attractive, so I may make the eye regions brighter. If their hair or dress is particularly appealing, I may choose to bring it out with increased saturation or modifications in texture or coloring.

These changes can be construed as unethical or irresponsible because they inaccurately represent who the person really is but rather present an idealistic, "fine-tuned" version. Thus, the image manipulations can be seen as dishonest. Moreover, by manipulating images to make people more "attractive," we reinforce common and arguably incorrect stereotypes about what beauty entails. For example, editing images to whiten people's teeth and remove their acne reinforces the idea that people who have yellow teeth or acne are not attractive, which is not necessarily the case. It does not appreciate people's natural physical appearance, which should not require editing to be appreciated.

Instead of attempting to edit the specific features of individuals to make them look more "attractive," I would instead rescope the project to focus more on improving the images' general quality, as well as editing the images to bring out the person in the image (and all of their features, rather than just the "good" ones). For example, if the person in the image is not centered, I may apply crops to center or make the person more visible, so that the viewer can more clearly see them. If the lighting is poor and the person's features are not visible, I could use manipulations to improve the lighting. I could also apply the image sharpening and denoising techniques mentioned above, which would make the image clearer. This would avoid the ethical problems mentioned earlier because the project no longer aims to change peoples' appearances to match some sort of stereotypical notion of beauty. In fact, by making the images overall look nicer and clearer, we could bring out the humanity and natural beauty of the individual in the image for the viewer to appreciate. This kind of computational photography could contribute toward combating the stereotypes mentioned earlier by illustrating humans in their rawest form.